Unit 3 - Calculus
3.2 Optimization and Linear Regression

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MSDS 520: Math and Stats Foundations for Data Science
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Outline

1. Why Optimize?
2. The Language of Optimization
3. The Derivative Test
4. The Second Derivative
5. Gradient Descent
6. The Gradient
7. Linear Regression
8. Way 1: Set the Partials to Zero
9. Way 2: Gradient Descent
10. Review

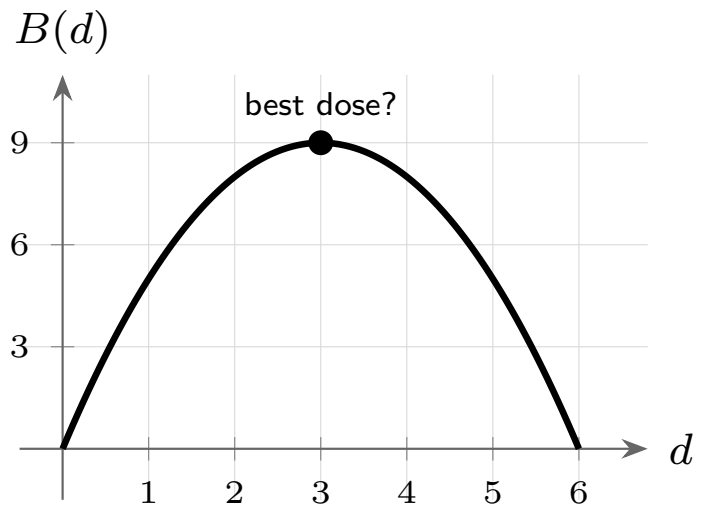
Why Optimize?

A Problem to Motivate Us

Choosing a dose

- More drug gives **more benefit**.
- More drug also gives **more side effects**, and those grow faster.
- A simplified model of the **net benefit** of a dose of d mg:

$$B(d) = 6d - d^2$$



Two questions this lecture answers

1. Where is the peak, **exactly**, without plotting? $\Rightarrow$ the **derivative test**
2. What if the formula is too complicated to solve? $\Rightarrow$ **gradient descent**

Where This Shows Up in Data Science

Optimization underneath the tools

- **Training a model.** A machine-learning model is **trained** by making its **loss** (how wrong its predictions are) as small as possible.
- **Fitting a line to data.** The best-fit line is the one with the smallest squared error. This lecture ends by building it.
- **Treatment planning.** Radiation therapy: maximize the dose to a tumor, minimize the dose to healthy tissue.
- **Hospital operations.** Staff schedules, bed assignments, supply orders.
- **Finance and engineering.** Portfolios (most return, least risk), aircraft wings (least drag).

Min or max: the same problem

Making f as **large** as possible is making $-f$ as **small** as possible. Machine learning always phrases it as a **minimum**.

What We Will Build

The plan

1. The **language**: objective function, search space, minimizer, local vs. global.
2. The **derivative test**: peaks and valleys are where $f'(x) = 0$.
3. The **second derivative**: how a curve bends, and how that tells a min from a max.
4. **Gradient descent**: when $f'(x) = 0$ cannot be solved, walk downhill instead.
5. The **learning rate**, when to stop, and the local-minimum trap.
6. The **gradient**: the partial derivatives (Unit 3-1) stacked into a vector that points uphill, so gradient descent works in two variables.
7. **Linear regression**: the loss $L(m, b)$, and the best line two ways: set the partials to zero, or descend.

Notebooks for this lecture: `U3-5_Optimize-1_DerivativeTest`, `U3-3_Derivatives-3_SecondDerivativeAndConcavity`, `U3-7_GradientDescent-1_Descent`, `U3-6_Multivariable-3`, `U3-7_Regression-1` to `-6`.

The Language of Optimization

Optimization Problems

The problem

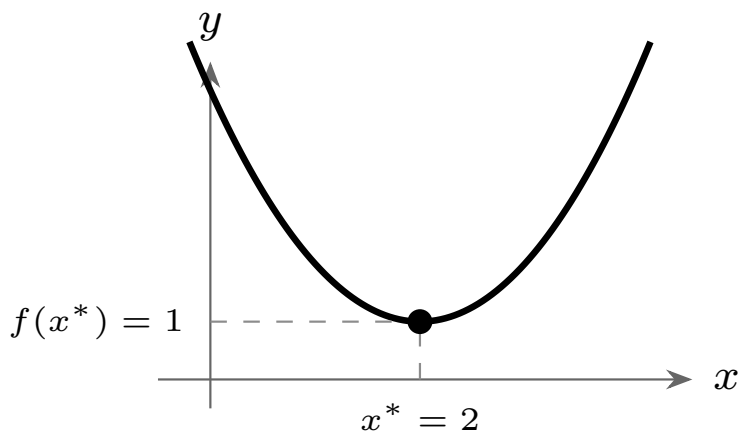
Find the smallest or largest value of a function $f(x)$ over a set of allowed inputs D:

$$\min_{x \in D} f(x) \qquad \max_{x \in D} f(x)$$

- f is the **objective function**: the score we care about.
- D is the **search space**: the inputs we are allowed to try.
- A minimum or a maximum is called an **extremum**.

Two answers: where, and how good

- The **minimizer** x^*: **where** the minimum happens.
- The **minimum value** $f(x^*)$: **how low** it gets.
- For $f(x) = (x-2)^2 + 1$: $x^* = 2$ and $f(x^*) = 1$.



Local vs. Global

Global minimum

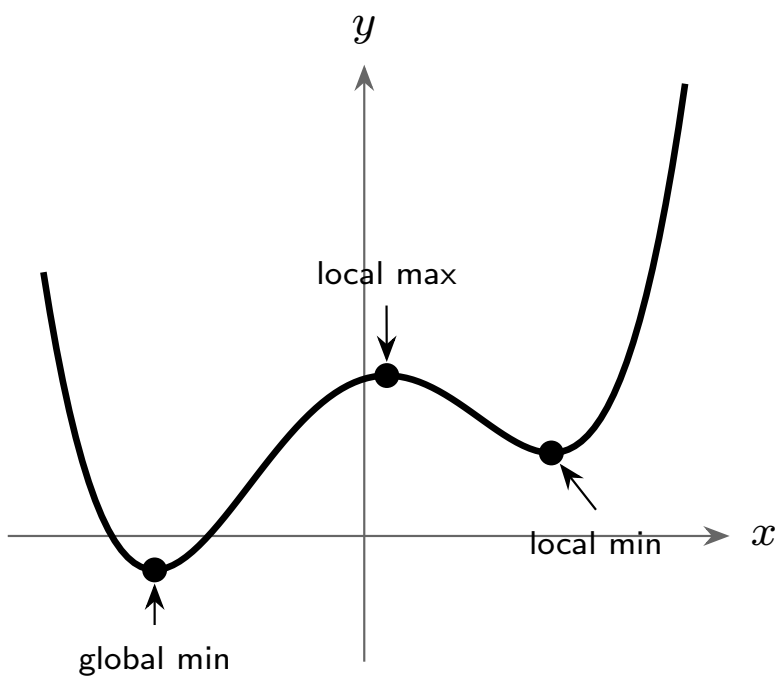
$f(x^*) \leq f(x)$ for **every** x in D: the lowest point anywhere.

Local minimum

$f(x^*) \leq f(x)$ for every x **near** x^*.

- You can't get better without first getting worse.
- Every global minimum is also local, not the other way around.

Local and global **maxima** are defined the same way, with $\geq$.



Why the Difference Matters

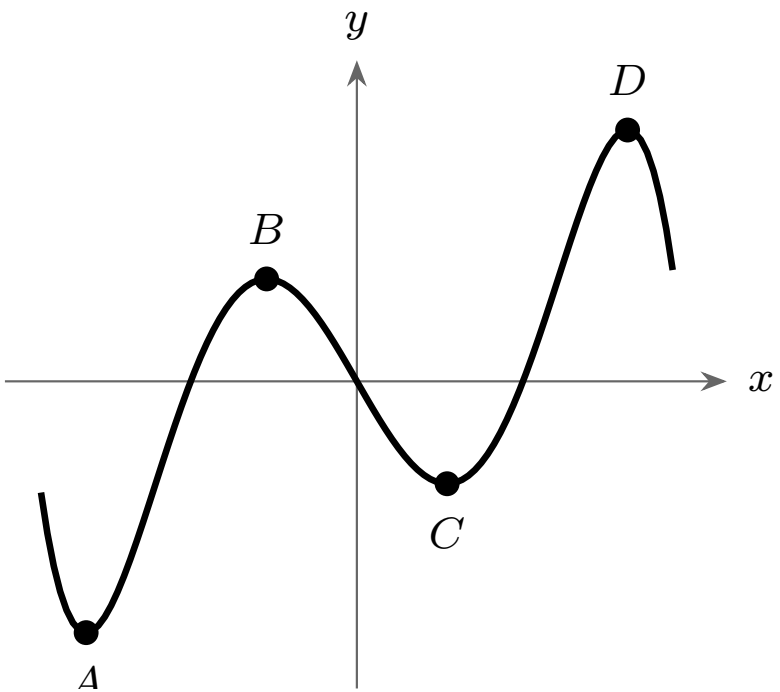
An algorithm walks in fog

- It sees only the ground **right under its feet**: the value and slope where it stands.
- At a local minimum, **every** small step goes uphill, so it looks exactly like the bottom.
- An algorithm can stop at a local minimum and report it as **the** answer.

Why not just plot it and look?

- With **one** input, a plot settles it.
- A machine-learning model has thousands to billions of inputs. There is no picture to look at.
- Most of the time we **cannot confirm** we found the global minimum. In practice the goal is a minimum that is **low enough**.

Exercise: Read the Graph



The search space is the window shown, $-3.5 \leq x \leq 3.5$.

Try it

1. Classify each of A, B, C, D as a local minimum or a local maximum.
2. Which point is the **global** minimum on this window? The global maximum?
3. Which points are local but **not** global? Why?
4. What do the tangent lines at all four points have in common?

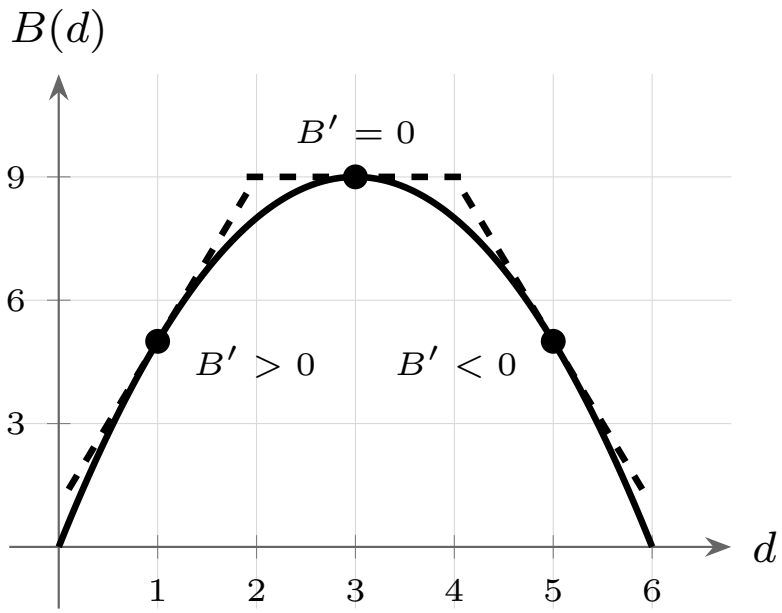
Exercise: Solution

Answers

1. Look at the neighbors of each point:
   - A and C sit at the bottom of a valley: every nearby point is higher. **Local minima.**
   - B and D sit at the top of a hill: every nearby point is lower. **Local maxima.**
2. A is the lowest point anywhere on the window: the **global minimum**. D is the highest: the **global maximum**.
3. B and C. B is the top of its own hill, but D is higher. C is the bottom of its own valley, but A is lower. Each is best only **nearby**.
4. All four tangent lines are **horizontal**: slope 0, so $f'(x) = 0$ at every one of them. That is the key to finding extrema with calculus, and it is the next section.

The Derivative Test

At a Peak, the Tangent Is Flat



Dashed: tangent lines at $d = 1, 3, 5$.

Follow the slope over the hill

- Left of the peak: rising, $B'(d) > 0$.
- At the peak: **flat**, $B'(d) = 0$.
- Right of the peak: falling, $B'(d) < 0$.

To turn from rising to falling, the slope has to pass through 0.

Critical point

A point c where $f'(c) = 0$. At a smooth max or min **inside** D, the point is always a critical point.

The Derivative Test

Algorithm: find the extrema of f on D

1. **Differentiate**: find $f'(x)$.
2. **Solve** $f'(x) = 0$. The solutions are the **critical points**: the candidates.
3. If D has **endpoints**, add them to the candidates.
4. **Evaluate** f at every candidate. The largest value is the max, the smallest is the min.

Why this works

- A smooth function can only turn around where its slope is 0.
- So instead of trying **infinitely** many x values, we check a **short list**.

Notebook `U3-5_Optimize-1_DerivativeTest`, §1 and §2

Worked: The Best Dose

The steps, for $B(d) = 6d - d^2$

1. Differentiate: $B'(d) = 6 - 2d$.
2. Solve: $6 - 2d = 0$, so $2d = 6$, so $d = 3$.
3. No endpoints yet: any dose is allowed.
4. Evaluate: $B(3) = 18 - 9 = 9$.

Is it a max? Check the neighbors

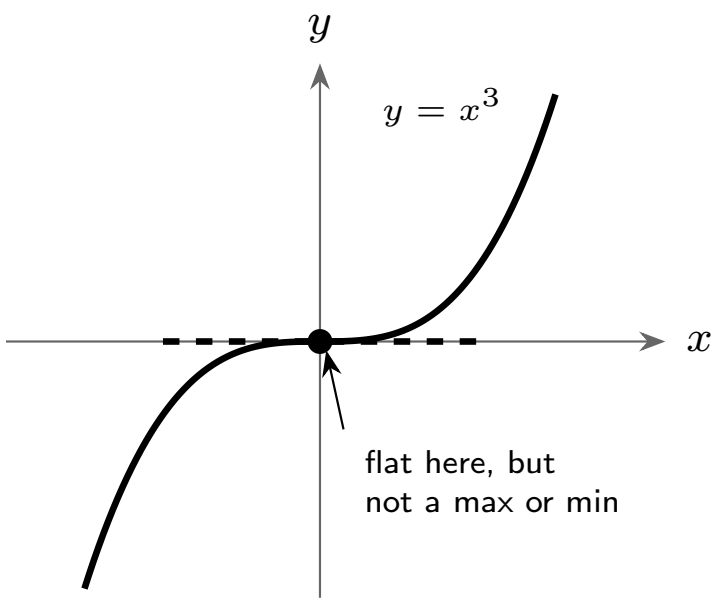
d	$B(d)$	$B'(d)$
2	8	$+2$
3	$\mathbf{9}$	0
4	8	-2

Lower on both sides: a **maximum**.

What it means

- Below 3 mg, more drug helps: $B'(d) > 0$.
- Above 3 mg, the side effects win: $B'(d) < 0$.
- 3 mg is the best tradeoff, with net benefit 9. We found it **without** plotting.

Flat Does Not Mean Extreme



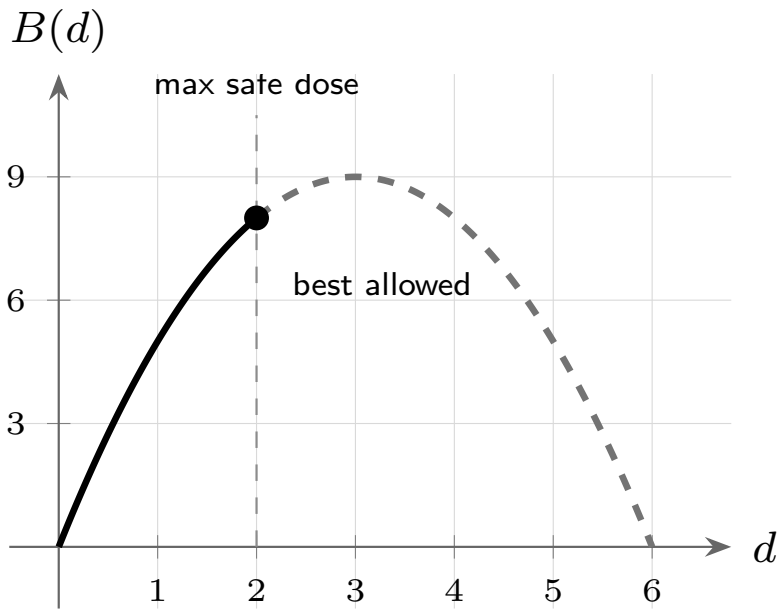
The example $f(x) = x^3$

- $f'(x) = 3x^2$, and $3x^2 = 0$ gives $x = 0$: a critical point.
- But $f(-0.1) < f(0) < f(0.1)$: the curve **rises on both sides**.
- It pauses flat, then keeps climbing. No turn, so no extremum.

Watch out

$f'(x) = 0$ gives **candidates**, not answers. Always check each one.

Watch the Endpoints



Solid: allowed doses. Dashed: not allowed.

Now the dose is capped at 2 mg

The search space is $D = [0, 2]$.

- $B'(d) = 6 - 2d > 0$ for every d in D: still rising.
- The critical point $d = 3$ is **outside** D.
- Candidates are the endpoints: $B(0) = 0$, $B(2) = 8$.
- Best: $d = 2$, where $B'(2) = 2 \neq 0$.

Watch out

With a restricted D (**constrained optimization**), the best value can sit on the **boundary**, where $f' \neq 0$.

Exercise: The Derivative Test

Try it

1. Find the critical point of $f(x) = x^2 - 8x + 20$. Is it a max or a min? What is the value there?
2. A clinic's daily profit, in hundreds of dollars, when it books q appointment slots is

$$P(q) = -2q^2 + 40q - 50.$$

   How many slots maximize profit, and what is the profit?
3. The clinic only has staff for **at most 5** slots, so $D = [0, 5]$. Now what is the best q?

Exercise: Solution

Answers

1. $f'(x) = 2x - 8 = 0$, so $x = 4$, and $f(4) = 16 - 32 + 20 = 4$. Neighbors: $f(3) = 5$ and $f(5) = 5$, both higher, so it is a **minimum**: the lowest value of f is 4, at $x = 4$.

2. $P'(q) = -4q + 40 = 0$, so $q = 10$, and $P(10) = -200 + 400 - 50 = 150$. Neighbors: $P(9) = P(11) = 148$, both lower, so it is a **maximum**. Book 10 slots for a profit of $\$15{,}000$ a day.

3. $P'(q) = -4q + 40 > 0$ for every q in $[0, 5]$: profit is **still rising** at the cap, and the critical point $q = 10$ is outside D. Check the endpoints:

   - $P(0) = -50$ and $P(5) = -50 + 200 - 50 = 100$.
   - Best: $q = 5$, profit $\$10{,}000$, at the **boundary**, where $P'(5) = 20 \neq 0$.

   Every extra slot up to the cap still adds profit; the staffing limit, not the market, is what stops us.

The Second Derivative

The Derivative of the Derivative

Definition

The **second derivative** $f''(x)$ is the derivative of $f'(x)$: differentiate twice.

$$f''(x) = \frac{d^2 f}{dx^2} = \frac{d}{dx}\left(\frac{df}{dx}\right)$$

- It measures how fast the **slope itself** is changing.

Example

$$f(x) = x^3 - 3x$$
$$f'(x) = 3x^2 - 3$$
$$f''(x) = 6x$$

In motion

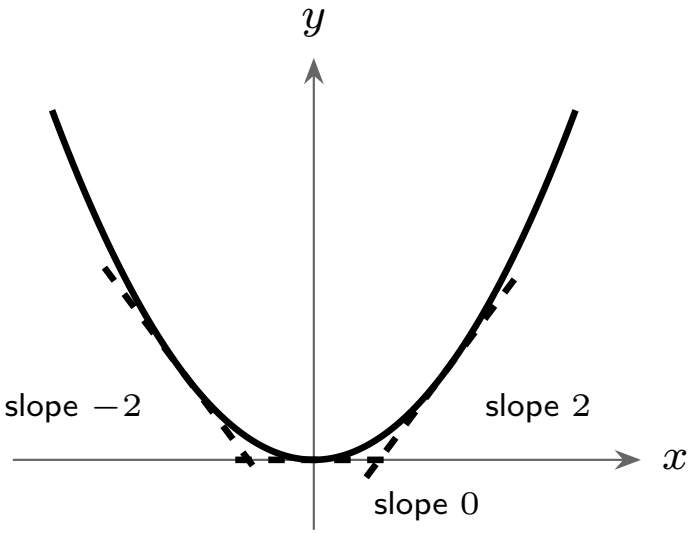
position	$x(t) = t^2$
velocity	$v(t) = x'(t) = 2t$
acceleration	$a(t) = x''(t) = 2$

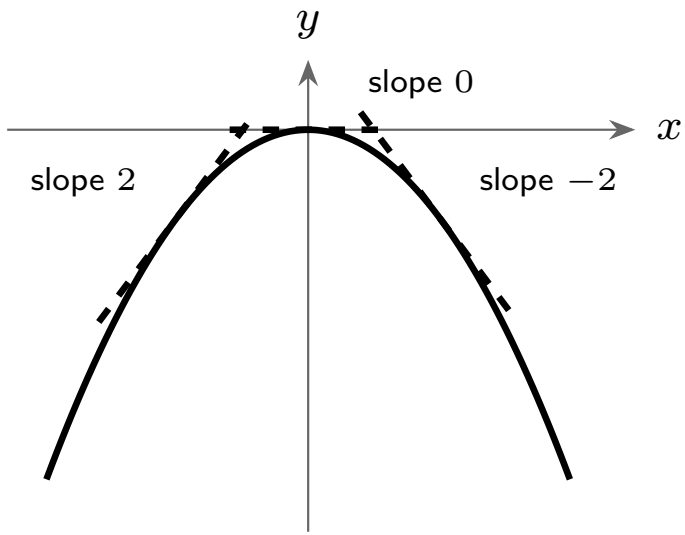
Acceleration is the second derivative of position: here, a steady speed-up.

Notebook `U3-3_Derivatives-3_SecondDerivativeAndConcavity`, §1 and §2

What f'' Measures: Bending

If the slope is **increasing**, the curve bends up. If it is **decreasing**, the curve bends down.





Concave up: a cup

Slopes increase, $-2 \to 0 \to 2$:
$f''(x) > 0$.

Concave down: a cap

Slopes decrease, $2 \to 0 \to -2$:
$f''(x) < 0$.

The Second Derivative Test

At a critical point c, where $f'(c) = 0$:

- $f''(c) > 0$: concave up, a cup. — c is a **local minimum**.
- $f''(c) < 0$: concave down, a cap. — c is a **local maximum**.
- $f''(c) = 0$: the test **says nothing**. — Check by hand.

Watch out: why $f''(c) = 0$ is inconclusive

All three of these have $f'(0) = 0$ **and** $f''(0) = 0$, yet they behave differently at 0:

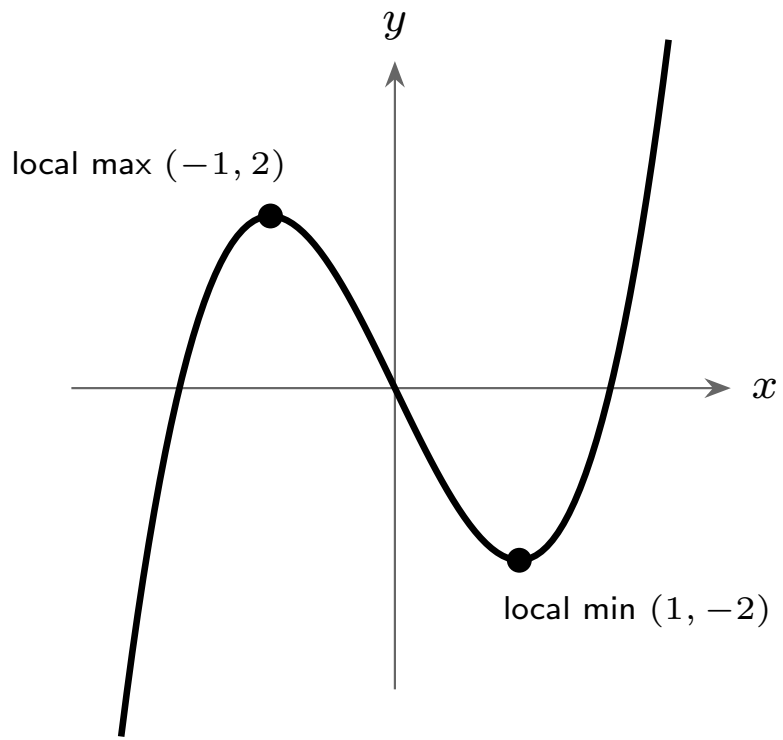
$f(x) = x^3$	$f(x) = x^4$	$f(x) = -x^4$
neither	minimum	maximum

Notebook `U3-5_Optimize-1_DerivativeTest`, §3

Worked: $f(x) = x^3 - 3x$

The steps

1. $f'(x) = 3x^2 - 3 = 3(x^2 - 1)$.
2. $3(x^2 - 1) = 0$ gives $x = -1$ and $x = 1$.
3. $f''(x) = 6x$:
   - $f''(-1) = -6 < 0$: **local max**, $f(-1) = -1 + 3 = 2$.
   - $f''(1) = 6 > 0$: **local min**, $f(1) = 1 - 3 = -2$.



Only local

Past $x = 2$ the curve climbs above 2 for good ($f(3) = 18$), so there is no global max.

Exercise: The Second Derivative Test

Try it

1. Find the critical points of $f(x) = 2x^3 - 6x^2 + 1$ and classify each with the second derivative test.
2. Show that the test is inconclusive for $f(x) = x^4$ at $x = 0$. Is $x = 0$ a min, a max, or neither?
3. A patient's temperature $T(t)$, t hours after admission, has $T'(6) = 0$ and $T''(6) < 0$. In plain words, what happened at hour 6?

Exercise: Solution

Answers

1. $f'(x) = 6x^2 - 12x = 6x(x-2)$, which is 0 at $x = 0$ and $x = 2$. Then $f''(x) = 12x - 12$:
   - $f''(0) = -12 < 0$: **local max**, with $f(0) = 1$.
   - $f''(2) = 24 - 12 = 12 > 0$: **local min**, with $f(2) = 16 - 24 + 1 = -7$.
2. $f'(x) = 4x^3$ and $f''(x) = 12x^2$, so $f'(0) = 0$ and $f''(0) = 0$: the test says nothing. By hand: $x^4 \geq 0 = f(0)$ for every x, so $x = 0$ is a **minimum**, in fact the global one.
3. $T'(6) = 0$: the temperature stopped changing for an instant. $T''(6) < 0$: the curve is a cap there. So the temperature **peaked** at hour 6 (a local max): the fever was rising before and falling after.

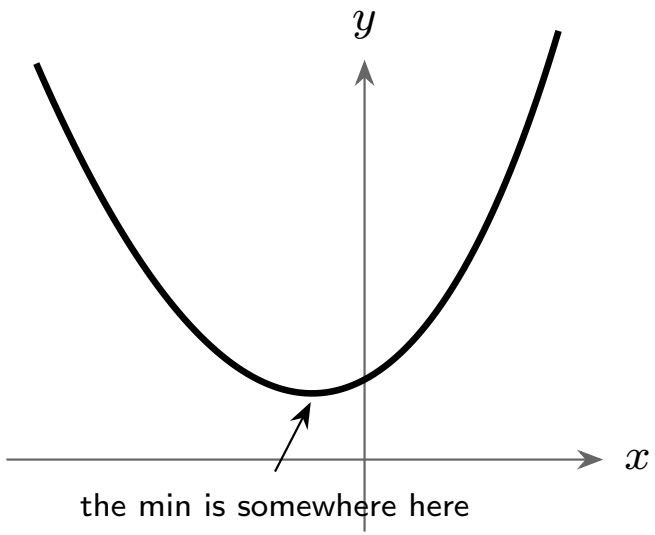
Gradient Descent

When You Cannot Solve $f'(x) = 0$

A harmless-looking function

$$f(x) = e^x + x^2 \quad \Rightarrow \quad f'(x) = e^x + 2x$$

- The minimum is where $e^x + 2x = 0$.
- No algebra isolates x in that equation.
- The derivative test **stalls at step 2**.



This is the normal case in machine learning

- A model's loss has thousands to billions of inputs, and $f' = 0$ has no formula for a solution.
- We need a method that only ever **evaluates** f' at a point, and never **solves** for where it is 0.

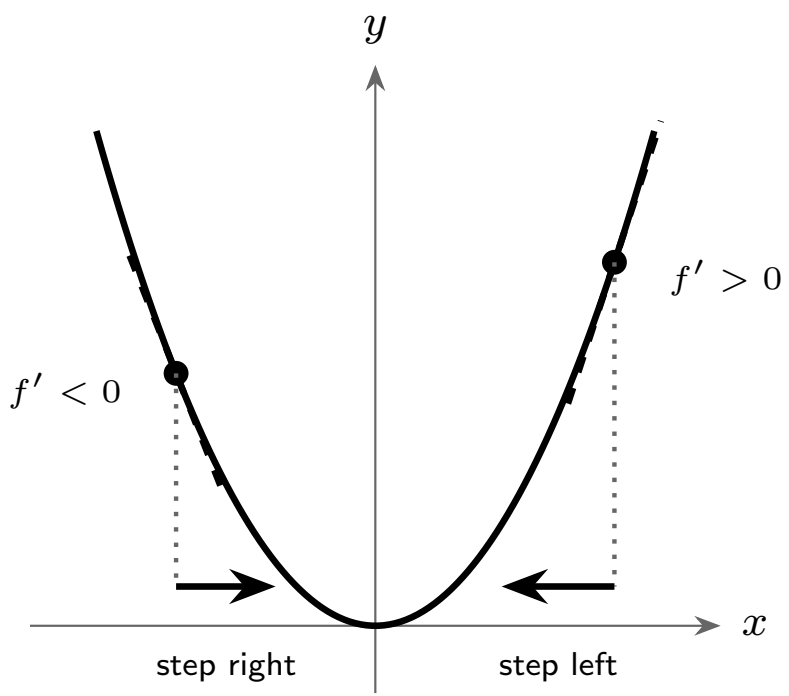
The Idea: Walk Downhill

Lost on a hillside in fog

1. Start somewhere: a **guess**.
2. Feel the slope under your feet.
3. Take a step **downhill**.
4. Repeat until the ground is flat.

Which way is downhill?

- $f'(x) > 0$: rising to the right, so move **left**.
- $f'(x) < 0$: falling to the right, so move **right**.
- Either way: move **against** the sign of $f'(x)$.



Dashed: tangent lines. Arrows: the step from each point, against the slope.

The Update Rule

Gradient descent, one step

$$x_{k+1} = x_k - \alpha\, f'(x_k)$$

Reading each piece

- x_k: where we stand now (step k). x_0 is the starting guess.
- $f'(x_k)$: the slope where we stand.
- α: the **learning rate**, a small positive number such as 0.1. It sets the step size. (The notebook writes η, eta.)
- The **minus sign**: step **against** the slope, which is downhill.

Two things to notice

- A steep slope gives a **big** step; a gentle slope gives a **small** one.
- At a minimum, $f'(x_k) = 0$, so $x_{k+1} = x_k$: the walk stops by itself.

Gradient Descent: the Algorithm

Algorithm: gradient descent

1. Choose a learning rate α, a maximum number of steps N, and a starting guess x_0.
2. For $k = 0, 1, \ldots, N-1$:
   - compute the slope $f'(x_k)$;
   - update $x_{k+1} = x_k - \alpha f'(x_k)$;
   - stop early if the step was tiny.
3. Return the last x.

When to stop

- After a fixed number of steps N.
- When the slope is tiny: $|f'(x_k)|$ below a small tolerance.
- When the steps are tiny: $|x_{k+1} - x_k|$ below a tolerance.

In practice: a tolerance, **plus** a cap N as a safety net.

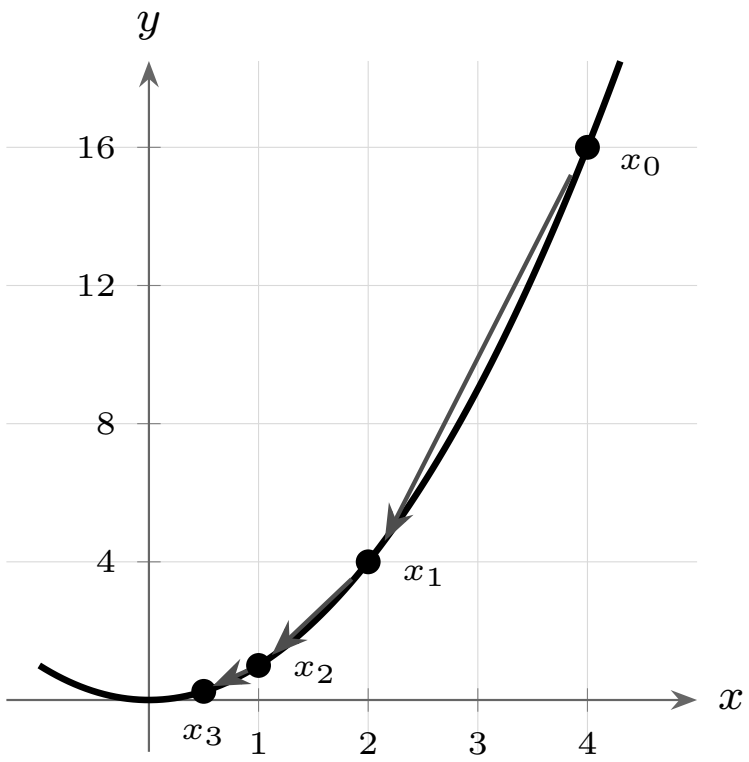
Worked: $f(x) = x^2$, $\alpha = 0.25$, $x_0 = 4$

Here $f'(x) = 2x$

k	x_k	$f'(x_k)$	$\alpha f'(x_k)$	x_{k+1}
0	4	8	2	2
1	2	4	1	1
2	1	2	0.5	0.5

What to notice

- Each step: $x_{k+1} = x_k - 0.25 \cdot 2x_k = \frac{1}{2}x_k$, so x **halves**.
- Steps shrink, $2, 1, 0.5$, as the slope flattens near the bottom.
- It heads for the true minimum $x = 0$ without ever solving $2x = 0$.



Notebook `U3-7_GradientDescent-1_Descent`, §1

Choosing the Learning Rate

For $f(x) = x^2$ the update is $x_{k+1} = x_k - \alpha \cdot 2x_k = (1 - 2\alpha)\, x_k$. Start at $x_0 = 4$:

Same function, same start, four learning rates

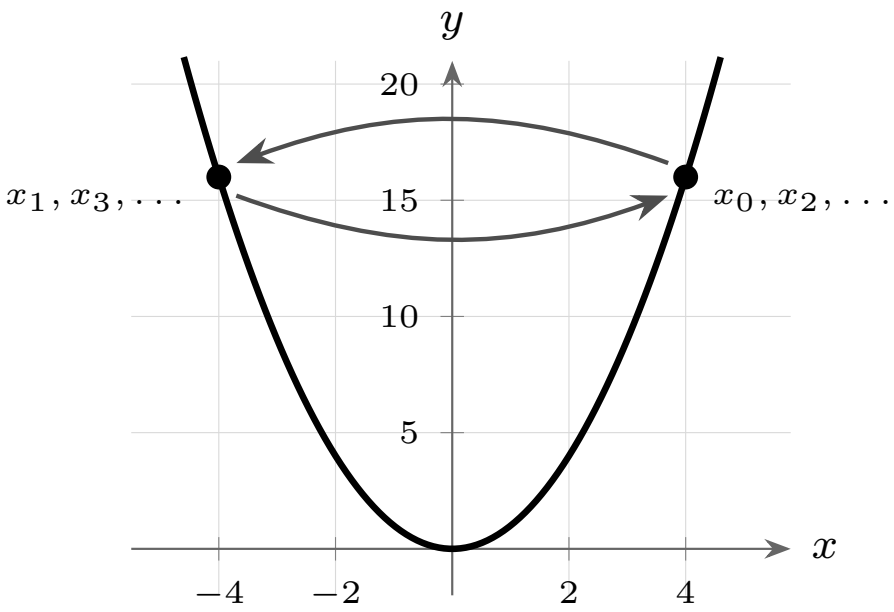
α	$1 - 2\alpha$	x_0	x_1	x_2	x_3	
0.05	0.9	4	3.6	3.24	2.916	**too small**: crawls
0.25	0.5	4	2	1	0.5	**good**: steady
1	-1	4	-4	4	-4	**too big**: bounces forever
1.25	-1.5	4	-6	9	-13.5	**too big**: blows up

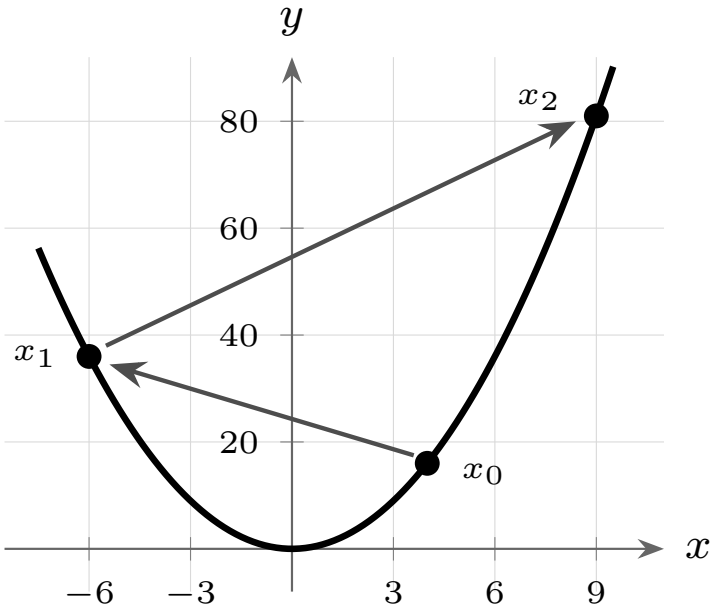
The tradeoff

- Too small: correct, but it may need thousands of steps.
- Too big: each step **overshoots** the bottom and lands on the far side, possibly higher than before.
- There is no universal right α: in practice you try a few and watch $f(x_k)$ fall.

Notebook `U3-7_GradientDescent-1_Descent`, §3

Too Big: Overshooting





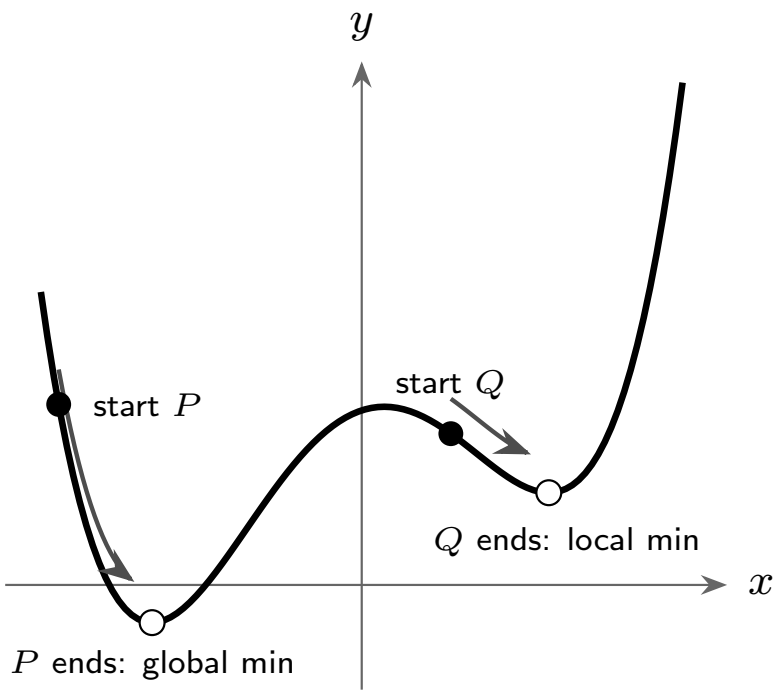
$\alpha = 1$: bounces forever

$4 \to -4 \to 4 \to \cdots$, never closer to 0.

$\alpha = 1.25$: blows up

$4 \to -6 \to 9$: each landing is **higher**.

The Local Minimum Trap



The start decides the end

- $f'(x) = 0$ at a local min too, so the walk stops there.
- Q starts above the shallow valley and never sees the deep one.

What people do about it

- **Restarts**: run from several starts, keep the lowest result.
- Smarter optimizers that build up speed to roll through small dips.

Notebook `U3-7_GradientDescent-1_Descent`, §4

Strengths and Weaknesses

Why it is used everywhere

- **Simple**: one line of arithmetic, repeated.
- Needs only to **evaluate** f' at a point, never to solve $f'(x) = 0$.
- Steps scale with the slope: big when far away, small near the bottom.
- Works the same with millions of inputs; this lecture takes it to two.

What can go wrong

- Gets **stuck** in local minima.
- A bad learning rate: too small crawls, too big bounces or blows up.
- Needs the derivative, as a formula or a numerical approximation.
- In very deep models the slopes can become huge or nearly zero, so the steps blow up or stall.

Gradient Descent in Python

Minimizing $f(x) = e^x + x^2$

```
import numpy as np

def df(x):                  # slope of f
    return np.exp(x) + 2*x

x = 1.0                     # starting guess x_0
alpha = 0.1                 # learning rate
for k in range(100):        # N = 100 steps
    x = x - alpha * df(x)

print(x)
# -0.3517337112491751
print(df(x))
# 5.60662627435704e-14
```

Reading the code

- The loop body **is** the update rule.
- We only ever **evaluate** `df(x)`.

Check the answer

`e-14` means $\times 10^{-14}$: the final slope is essentially 0, so the ground is flat. We found where $e^x + 2x = 0$ without solving it.

Notebook `U3-7_GradientDescent-1_Descent`, §2

Exercise: Gradient Descent by Hand

Try it

Let $f(x) = (x-3)^2$, so $f'(x) = 2(x-3)$ by the chain rule.

1. With $\alpha = 0.25$ and $x_0 = 1$, compute x_1 and x_2.
2. Where is the true minimum? How far from it is each of x_0, x_1, x_2? What pattern do you see?
3. Repeat part 1 with $\alpha = 1$. What goes wrong?

Exercise: Solution

Answers

1. Two steps with $\alpha = 0.25$:
   - $f'(1) = 2(1-3) = -4$, so $x_1 = 1 - 0.25(-4) = 1 + 1 = 2$.
   - $f'(2) = 2(2-3) = -2$, so $x_2 = 2 - 0.25(-2) = 2 + 0.5 = 2.5$.

   The slope was negative, so each step moved **right**, toward the bottom.
2. The minimum is at $x = 3$ (where $f'(x) = 0$). Distances: 2, then 1, then 0.5. Each step **halves** the distance, so the walk closes in on 3 but the steps keep shrinking.
3. With $\alpha = 1$:
   - $x_1 = 1 - 1 \cdot (-4) = 5$, then $f'(5) = 4$, so $x_2 = 5 - 1 \cdot 4 = 1$.

   It jumps from 2 left of the minimum to 2 right of it and back: it **bounces** between 1 and 5 forever. The learning rate is too big.

The Gradient

Exercise: Partial Derivatives

Try it: a warm-up on partial derivatives (Unit 3-1)

1. For $f(x, y) = x^3 + 4xy - y^2$, find $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$, then evaluate both at $(1, 2)$.
2. A dose D (mg) spreads through a volume V (L) of body fluid, giving the concentration $C(D, V) = \dfrac{D}{V}$.
   - Find $\dfrac{\partial C}{\partial D}$ and $\dfrac{\partial C}{\partial V}$.
   - Evaluate both at $D = 500$ and $V = 50$, with units, and say what each one means.

Exercise: Solution

Answers

1. $\dfrac{\partial f}{\partial x} = 3x^2 + 4y$ (freeze y; $4xy$ is the constant $4y$ times x), $\dfrac{\partial f}{\partial y} = 4x - 2y$ (freeze x).
   - At $(1, 2)$: $\frac{\partial f}{\partial x} = 3 + 8 = 11$ and $\frac{\partial f}{\partial y} = 4 - 4 = 0$.
   - Meaning: stepping in x, the surface climbs steeply (slope 11); stepping in y, it is momentarily **flat**.
2. Write $C = D \cdot V^{-1}$. Here $C = 500/50 = 10$ mg/L.
   - $\dfrac{\partial C}{\partial D} = \dfrac{1}{V} = \dfrac{1}{50} = 0.02$ (mg/L) per mg: each extra mg of dose raises the concentration by 0.02 mg/L.
   - $\dfrac{\partial C}{\partial V} = -\dfrac{D}{V^2} = -\dfrac{500}{2500} = -0.2$ (mg/L) per L: the same dose in a patient with 1 L more fluid gives about 0.2 mg/L **less**.

The Gradient Vector

Definition

The **gradient** of $f(x, y)$ stacks its two partials into a column vector:

$$\nabla f(x, y) = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \end{bmatrix}$$

- ∇ is read "del". It is a Unit 2 vector at **every** point, with a **direction** and a **length**: the next slide reads both.

$f(x, y) = x^2 + y^2$

$\frac{\partial f}{\partial x} = 2x$ (y^2 is a constant), $\frac{\partial f}{\partial y} = 2y$ (x^2 is a constant):

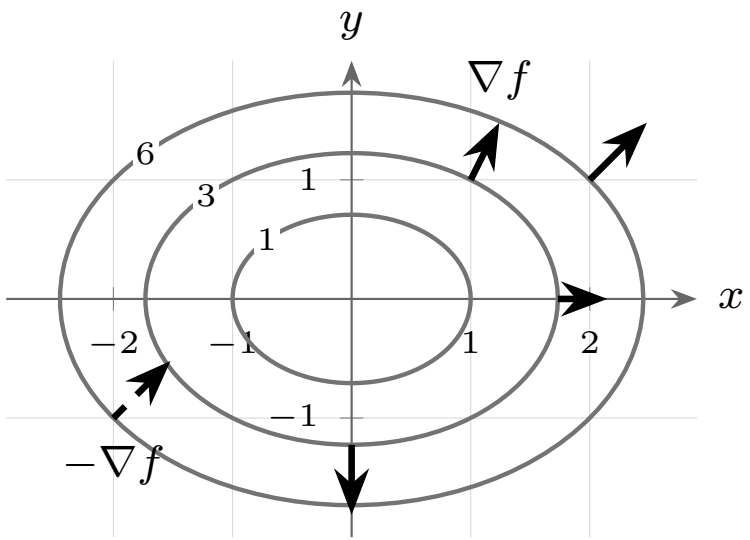
$$\nabla f = \begin{bmatrix} 2x \\ 2y \end{bmatrix}, \qquad \nabla f(1, 2) = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$g(x, y) = x^2 y + 3y$

$\frac{\partial g}{\partial x} = 2xy$ (y rides along), $\frac{\partial g}{\partial y} = x^2 + 3$ (x^2 rides along):

$$\nabla g = \begin{bmatrix} 2xy \\ x^2 + 3 \end{bmatrix}, \qquad \nabla g(1, 2) = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

The Gradient Points Uphill



Contours of $f = x^2 + 2y^2$ at heights $1, 3, 6$.
Arrows drawn at 0.12 times their length.

What the arrows say

- **Direction**: ∇f points the way of **steepest climb**.
- The dashed arrow, $-\nabla f$, points **steepest downhill**.
- **Length**: how steep. Arrows are longer where contours crowd.
- Each arrow crosses its contour at a **right angle**.
- At a flat point (bottom of a bowl, top of a hill, center of a saddle), $\nabla f = \bar{0}$.

Notebook `U3-6_Multivariable-3_GradientField`, §1 to §4

Why Uphill? The Dot Product

The slope in any direction

Walk in the direction of a unit vector $\hat{u}$. The height changes at the rate

$$D_{\hat{u}} f = \hat{u} \cdot \nabla f = \|\nabla f\| \cos\theta,$$

the **directional derivative**. It is the Unit 2-2 dot product, with $\|\hat{u}\| = 1$ and θ the angle between $\hat{u}$ and ∇f.

Reading $\cos\theta$

θ	$D_{\hat{u}} f$	**direction**
$0°$	$+\|\nabla f\|$	along ∇f: the fastest climb
$90°$	0	along the contour: level ground
$180°$	$-\|\nabla f\|$	along $-\nabla f$: the fastest drop

Check: $f = x^2 + y^2$ at $(1, 2)$, where $\nabla f = \left[\begin{smallmatrix} 2 \\ 4 \end{smallmatrix}\right]$

Along $\hat{\imath}$ the slope is $2 = \partial f/\partial x$. Steepest climb: $\|\nabla f\| = \sqrt{20} \approx 4.47$.

Gradient Descent in Two Variables

From one variable to two

$$x_{k+1} = x_k - \alpha f'(x_k) \qquad \longrightarrow \qquad \begin{bmatrix} x_{k+1} \\ y_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \end{bmatrix} - \alpha \nabla f(x_k, y_k)$$

Same idea: step **against** the slope. Now the gradient says which way is downhill.

The algorithm

1. Choose a learning rate α, a number of steps N, and a start (x_0, y_0).
2. Compute the gradient at the current point.
3. Step against it: subtract $\alpha \nabla f$ from the current point.
4. Repeat 2 and 3 until ∇f is nearly $\bar{0}$ or N steps are done.

Same cautions as in one variable: α too small crawls; too big overshoots or blows up. It can still stop in a local minimum.

Notebook `U3-7_GradientDescent-1_Descent`, §1 and §3 (one variable)

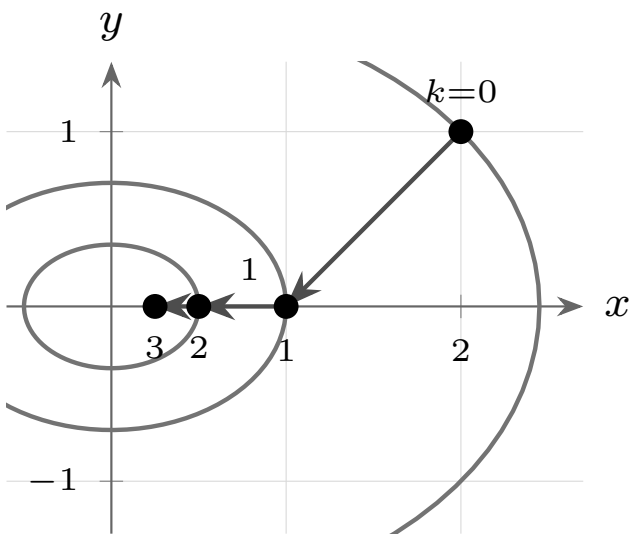
Worked: Descent on $x^2 + 2y^2$

Start $(2, 1)$, $\alpha = 0.25$

$\nabla f = \left[\begin{smallmatrix} 2x \\ 4y \end{smallmatrix}\right]$. Step 1:

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} - 0.25 \begin{bmatrix} 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

k	(x_k, y_k)	∇f	f
0	$(2, 1)$	$\left[\begin{smallmatrix} 4 \\ 4 \end{smallmatrix}\right]$	6
1	$(1, 0)$	$\left[\begin{smallmatrix} 2 \\ 0 \end{smallmatrix}\right]$	1
2	$(0.5, 0)$	$\left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right]$	0.25
3	$(0.25, 0)$	$\left[\begin{smallmatrix} 0.5 \\ 0 \end{smallmatrix}\right]$	0.0625



Labels are step numbers k.

What happened

f drops at every step. y reaches 0 in one step; after that x halves each step, heading for the minimum at $(0, 0)$.

Exercise: The Gradient

Try it

Let $f(x, y) = x^2 + 3y^2 + xy$.

1. Find $\nabla f(x, y)$.
2. Evaluate it at $(1, -1)$.
3. Take one gradient descent step from $(1, -1)$ with $\alpha = 0.1$.
4. Compute f before and after the step. Did it go down?

Exercise: Solution

Answers

1. In x, the term xy is the constant y times x; in y, it is x times y. So
$$\nabla f = \begin{bmatrix} 2x + y \\ 6y + x \end{bmatrix}.$$
2. $\nabla f(1, -1) = \begin{bmatrix} 2 - 1 \\ -6 + 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -5 \end{bmatrix}$.
3. $\begin{bmatrix} 1 \\ -1 \end{bmatrix} - 0.1 \begin{bmatrix} 1 \\ -5 \end{bmatrix} = \begin{bmatrix} 0.9 \\ -0.5 \end{bmatrix}$.
4. Before: $f(1, -1) = 1 + 3 - 1 = 3$. After: $f(0.9, -0.5) = 0.81 + 0.75 - 0.45 = 1.11$.

Meaning: the step went downhill, from 3 to 1.11. Most of the move was in y, because the surface is much steeper in y there (5 against 1).

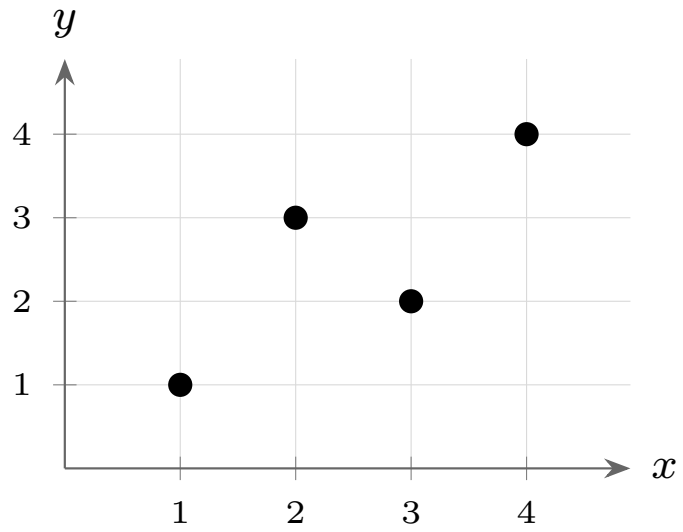
Linear Regression

Four Patients: Which Line Fits Best?

Four patients

patient	dose x (mg)	relief y
1	1	1
2	2	3
3	3	2
4	4	4

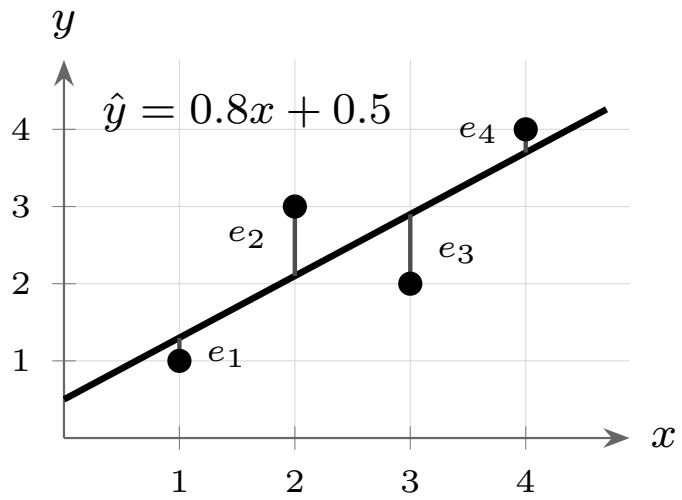
Relief is a symptom score: higher is better.



The question

- The trend is upward, but not a perfect line. **Which** line $\hat{y} = mx + b$ fits best?
- "Best" will be one number, the **loss**, that depends on **two** choices: the slope m and the intercept b.
- Minimizing a function of two variables: **partial derivatives** (Unit 3-1) and the **gradient**.

The Model and Its Residuals



x_i	1	2	3	4
y_i	1	3	2	4
$\hat{y}_i$	1.3	2.1	2.9	3.7
e_i	-0.3	0.9	-0.9	0.3

The model

$$\hat{y} = mx + b$$

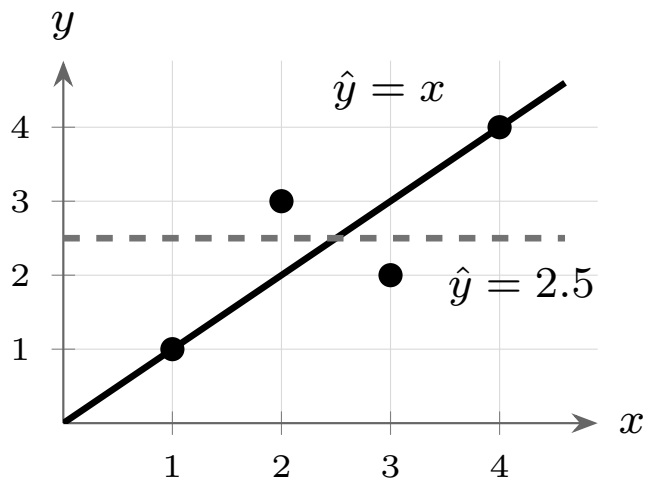
- m is the slope, b the intercept.
- The hat means **predicted**. It is not the unit-vector hat of Unit 2.
- The notebooks write the same line as $\hat{y} = b_0 + b_1 x$.

Residuals: $e_i = y_i - \hat{y}_i$

Actual minus predicted: the vertical gap. Positive for a point above the line, negative below. The table lists them for this candidate line.

Notebook `U3-7_Regression-1_FittingALine`, §2 and §3

Why Square the Residuals?



Solid: $\hat{y} = x$. Dashed: $\hat{y} = 2.5$.

Adding raw residuals fails

Positive and negative misses **cancel**: both lines total 0, a "perfect" score, yet the flat line plainly fits worse.

Squaring fixes it

Every miss counts as positive, and **big** misses cost more: a miss of 2 costs 4, a miss of 0.5 only 0.25.

Two lines, same data

line	**residuals**	$\sum e_i$	$\sum e_i^2$
$\hat{y} = x$	$0,\ 1,\ -1,\ 0$	0	2
$\hat{y} = 2.5$	$-1.5,\ 0.5,\ -0.5,\ 1.5$	0	5

Notebook `U3-7_Regression-1_FittingALine`, §3

The Loss Function

Mean squared error (MSE)

$$L(m, b) = \frac{1}{n} \sum_{i=1}^{n} \big(y_i - (mx_i + b)\big)^2$$

The average squared residual; smaller is better. For our four patients:

$$L(m, b) = \tfrac{1}{4}\Big[(1 - m - b)^2 + (3 - 2m - b)^2 + (2 - 3m - b)^2 + (4 - 4m - b)^2\Big]$$

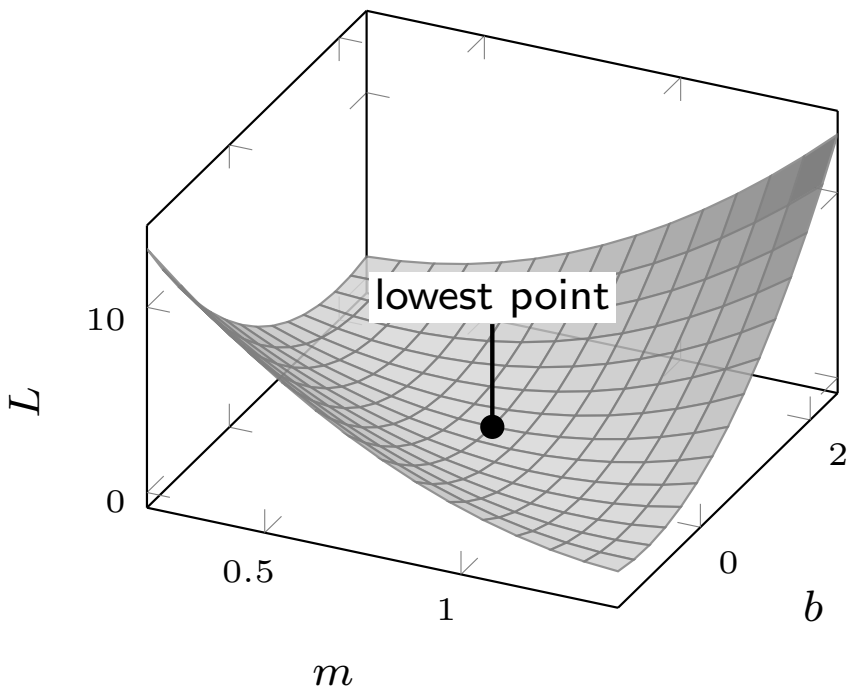
Scoring three lines

line	L
$\hat{y} = 2.5$	$5/4 = 1.25$
$\hat{y} = x$	$2/4 = 0.5$
$\hat{y} = 0.8x + 0.5$	$1.8/4 = 0.45$

The key shift

- The data x_i, y_i are **fixed numbers**.
- The only things we choose are m and b.
- So L is a **function of two variables**, and the best line is its **lowest point**.

The Loss Is a Bowl



What the surface says

- **One** lowest point: that pair (m, b) is the best line. The next two sections find it, two ways.
- **No local-minimum trap**: for a line with squared error, the loss is always a bowl with one bottom.
- A long, narrow **valley**: raise m and lower b together and the line pivots near the middle of the data, so the loss barely changes.

Notebook `U3-7_Regression-3_ThreeWays`, §1

Way 1: Set the Partials to Zero

The Partials of the Loss

One squared residual, by the chain rule (Unit 3-1)

Inner $u = y_i - mx_i - b$ (data are constants); outer u^2, derivative $2u$.

$$\frac{\partial}{\partial m} u^2 = 2u \cdot \frac{\partial u}{\partial m} = 2(y_i - mx_i - b)(-x_i)$$

$$\frac{\partial}{\partial b} u^2 = 2u \cdot \frac{\partial u}{\partial b} = 2(y_i - mx_i - b)(-1)$$

Average over all n points

$$\frac{\partial L}{\partial m} = -\frac{2}{n}\sum_{i=1}^{n} x_i\,(y_i - mx_i - b), \qquad \frac{\partial L}{\partial b} = -\frac{2}{n}\sum_{i=1}^{n}(y_i - mx_i - b)$$

In words: each is a total of the residuals e_i, weighted by x_i for m and unweighted for b.

The Normal Equations

At the bottom of the bowl both slopes are zero: the derivative test $f'(x) = 0$ from earlier, once per variable.

From $\partial L/\partial b = 0$

Divide by $-\frac{2}{n}$ and split the sum (b added n times is nb):

$$\sum y_i - m \sum x_i - nb = 0$$

$$m \sum x_i + b\, n = \sum y_i$$

From $\partial L/\partial m = 0$

Divide by $-\frac{2}{n}$ and split the sum:

$$\sum x_i y_i - m \sum x_i^2 - b \sum x_i = 0$$

$$m \sum x_i^2 + b \sum x_i = \sum x_i y_i$$

The normal equations: a 2×2 linear system (Unit 2-1)

$$\left[\begin{array}{cc|c} \sum x_i^2 & \sum x_i & \sum x_i y_i \\ \sum x_i & n & \sum y_i \end{array}\right]$$

Two linear equations in m and b. Every $\sum$ is just a number from the data.

Solving the Running Example

The sums

x_i	y_i	x_i^2	$x_i y_i$
1	1	1	1
2	3	4	6
3	2	9	6
4	4	16	16
10	**10**	**30**	**29**

and $n = 4$.

The system, by elimination

$$30m + 10b = 29$$
$$10m + 4b = 10$$

- $3\times$ (row 2): $30m + 12b = 30$.
- Subtract row 1: $2b = 1$, so $b = 0.5$.
- Into row 2: $10m + 2 = 10$, so $m = 0.8$.

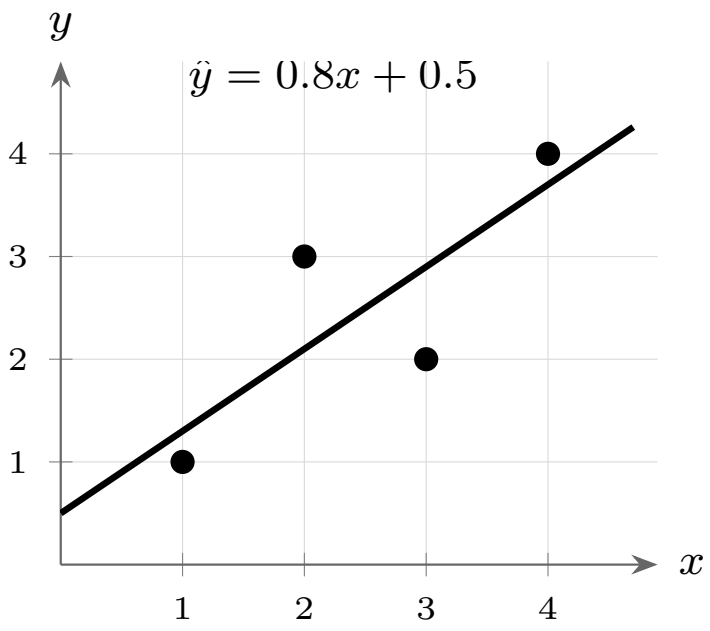
Or with the Unit 2-5 inverse

$\det = 30 \cdot 4 - 10 \cdot 10 = 20 \neq 0$, so there is **exactly one** best line:

$$\begin{bmatrix} m \\ b \end{bmatrix} = \frac{1}{20} \begin{bmatrix} 4 & -10 \\ -10 & 30 \end{bmatrix} \begin{bmatrix} 29 \\ 10 \end{bmatrix} = \frac{1}{20} \begin{bmatrix} 16 \\ 10 \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.5 \end{bmatrix}$$

Notebook `U3-7_Regression-3_ThreeWays`, §3

What the Best Line Says



Reading m and b

- The best line is $\hat{y} = 0.8x + 0.5$: the candidate we drew earlier.
- **Slope** $m = 0.8$: each extra mg of dose raises predicted relief by 0.8 points.
- **Intercept** $b = 0.5$: predicted relief at dose 0. No patient had dose 0, so read it with care.
- **Prediction**: a 2.5 mg dose gives $0.8(2.5) + 0.5 = 2.5$.

Checks

- Its loss, 0.45, beats both earlier lines (0.5 and 1.25), and every other line.
- Its residuals $-0.3,\ 0.9,\ -0.9,\ 0.3$ sum to 0: exactly what $\partial L/\partial b = 0$ says.

Matrix Form: What Unit 2-5 Promised

Stack the data

$$X = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix}, \qquad \bar{y} = \begin{bmatrix} 1 \\ 3 \\ 2 \\ 4 \end{bmatrix}, \qquad X \begin{bmatrix} m \\ b \end{bmatrix} = \begin{bmatrix} m+b \\ 2m+b \\ 3m+b \\ 4m+b \end{bmatrix}$$

Column 1 holds the x_i; column 2 is all ones, to carry b. $X \left[\begin{smallmatrix} m \\ b \end{smallmatrix}\right]$ lists the four predictions.

The normal equations are $X^\top X \left[\begin{smallmatrix} m \\ b \end{smallmatrix}\right] = X^\top \bar{y}$

$$X^\top X = \begin{bmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & n \end{bmatrix} = \begin{bmatrix} 30 & 10 \\ 10 & 4 \end{bmatrix}, \qquad X^\top \bar{y} = \begin{bmatrix} \sum x_i y_i \\ \sum y_i \end{bmatrix} = \begin{bmatrix} 29 \\ 10 \end{bmatrix}$$

Exactly the system we just solved, so $\left[\begin{smallmatrix} m \\ b \end{smallmatrix}\right] = (X^\top X)^{-1} X^\top \bar{y}$. More features just add columns to X.

The notebook puts the ones column first, since it writes b_0 first.

Notebook `U3-7_Regression-4_MatrixForm`, §1 to §3

Exercise: Fit a Line

Try it

Three patients in physical therapy: x is weeks of therapy, y is an improvement score.

$$(0, 1), \qquad (1, 2), \qquad (2, 6)$$

1. Compute n, $\sum x_i$, $\sum y_i$, $\sum x_i^2$, and $\sum x_i y_i$.
2. Write the two normal equations.
3. Solve for m and b.
4. Find the three residuals and check that they sum to 0.
5. Say in words what the slope means.

Exercise: Solution

Answers

1. $n = 3$, $\sum x_i = 3$, $\sum y_i = 9$, $\sum x_i^2 = 0 + 1 + 4 = 5$, $\sum x_i y_i = 0 + 2 + 12 = 14$.
2. $5m + 3b = 14$ and $3m + 3b = 9$.
3. Subtract: $2m = 5$, so $m = 2.5$. Then $3(2.5) + 3b = 9$, so $3b = 1.5$ and $b = 0.5$.
   Best line: $\hat{y} = 2.5x + 0.5$. (And $\det = 5 \cdot 3 - 3 \cdot 3 = 6 \neq 0$, so it is the only one.)
4. Predictions $0.5,\ 3,\ 5.5$; residuals $0.5,\ -1,\ 0.5$; sum 0. ✓
5. Each extra week of therapy adds about 2.5 points of predicted improvement.

Way 2:
Gradient Descent

Gradient Descent on the Loss

The update

$$\begin{bmatrix} m \\ b \end{bmatrix} \leftarrow \begin{bmatrix} m \\ b \end{bmatrix} - \alpha \nabla L, \qquad \nabla L = \begin{bmatrix} \partial L/\partial m \\ \partial L/\partial b \end{bmatrix} = \begin{bmatrix} -\frac{2}{n}\sum x_i e_i \\ -\frac{2}{n}\sum e_i \end{bmatrix}$$

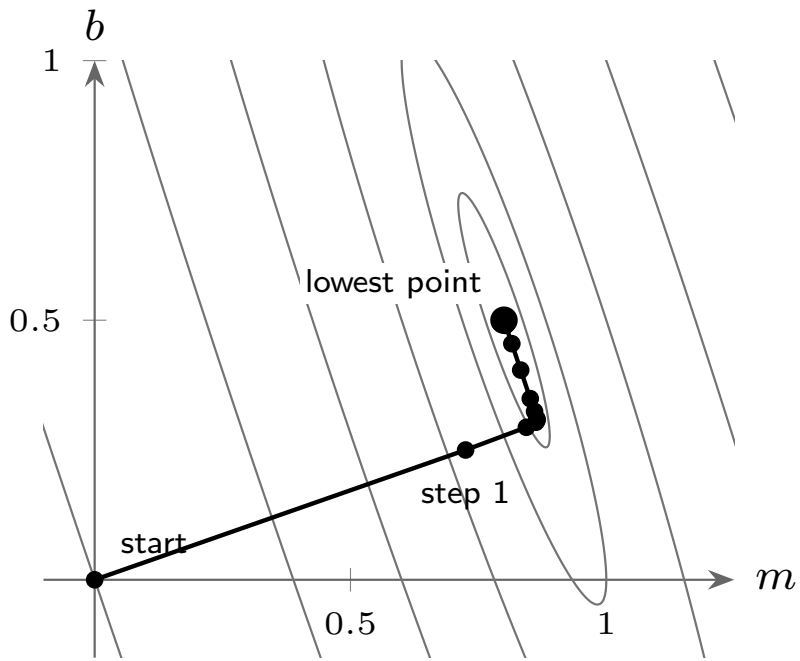
One step from $(m, b) = (0, 0)$ with $\alpha = 0.05$

- The line $\hat{y} = 0$ predicts 0, so the residuals are just the y_i: $1, 3, 2, 4$.
- $\dfrac{\partial L}{\partial m} = -\dfrac{2}{4}(1 \cdot 1 + 2 \cdot 3 + 3 \cdot 2 + 4 \cdot 4) = -\dfrac{1}{2}(29) = -14.5$
- $\dfrac{\partial L}{\partial b} = -\dfrac{2}{4}(1 + 3 + 2 + 4) = -\dfrac{1}{2}(10) = -5$
- New point: $m = 0 - 0.05(-14.5) = 0.725$ and $b = 0 - 0.05(-5) = 0.25$.

Meaning

Both partials are negative: raising m and b lowers the loss, so the step raises both. The loss falls from 7.5 to about 0.65 in one step.

The Path to the Bottom



Contours of $L(m, b)$; dots are steps.

$\alpha = 0.05$, from $(0, 0)$

k	(m_k, b_k)	L
0	$(0, 0)$	7.5
1	$(0.725, 0.250)$	0.6484
2	$(0.844, 0.294)$	0.4618
10	$(0.860, 0.324)$	0.4552
100	$(0.815, 0.455)$	0.4503
1000	$(0.800, 0.500)$	0.4500

What happened

- Two steps drop straight down the steep wall of the valley. Then a slow crawl along its narrow floor.
- It ends at $(0.8, 0.5)$: the **same line** as way 1.

Notebook `U3-7_Regression-3_ThreeWays`, §2

Why Gradient Descent If We Have a Formula?

The formula is a special case

- It exists because a **line** with **squared error** has partials that are **linear** in m and b, so setting them to 0 gives a linear system.
- Change the model and that breaks. **Logistic regression** (yes/no outcomes, such as readmitted or not) and **neural networks** have partials that no algebra can set to zero and solve.

Gradient descent only needs to **evaluate** the gradient

- No equation to solve: compute ∇L where you stand, step, repeat.
- It works for any loss you can differentiate, with 2 coefficients or 2 billion.
- Every neural network, including the language models behind chatbots, is trained this way.

More Than One Feature

Two features

Predict systolic blood pressure from age x_1 and BMI x_2:

$$\hat{y} = m_1 x_1 + m_2 x_2 + b, \qquad L(m_1, m_2, b) = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2$$

Both ways still work

- **Way 1**: three partials, all set to 0, give a 3×3 linear system. X gains a column, and $(X^\top X)^{-1} X^\top \bar{y}$ still holds.
- **Way 2**: ∇L now has 3 entries. Same update: subtract $\alpha \nabla L$ from the column of coefficients.
- **In general**, k features means $k+1$ coefficients, $k+1$ partials, and a gradient with $k+1$ entries. Nothing else changes.

Notebook `U3-7_Regression-5_MultipleFeatures`, §1 to §3; `-6_RealEstate` on real data

Linear Regression in Python

Ways 1 and 2, by hand in NumPy

```
import numpy as np
x = np.array([1, 2, 3, 4])
y = np.array([1, 3, 2, 4])
# Way 1: solve the normal equations
A = np.array([[np.sum(x*x), np.sum(x)],
              [np.sum(x),   len(x)]])
r = np.array([np.sum(x*y), np.sum(y)])
np.linalg.solve(A, r)
# array([0.8, 0.5])
# Way 2: gradient descent
m, b, alpha = 0.0, 0.0, 0.05
for k in range(1000):
    e = y - (m*x + b)           # residuals
    dm = -2 * np.mean(x * e)    # dL/dm
    db = -2 * np.mean(e)        # dL/db
    m, b = m - alpha*dm, b - alpha*db
print(round(m, 4), round(b, 4))
# 0.8 0.5
```

Way 3: the library

```
from sklearn.linear_model \
    import LinearRegression
X = x.reshape(-1, 1)
model = LinearRegression()
model.fit(X, y)
model.coef_, model.intercept_
# 0.8 and 0.5, up to rounding
```

Notes

- `np.mean` is the $\frac{1}{n}\sum$.
- `reshape(-1, 1)`: one row per patient, one column per feature.
- All three give the same line.

Notebook `U3-7_Regression-3`, §2 to §4

Exercise: One Step of Gradient Descent

Try it

Same four patients: $(1, 1)$, $(2, 3)$, $(3, 2)$, $(4, 4)$. Start at the line $\hat{y} = 0.5x + 0.5$, so $(m, b) = (0.5, 0.5)$, and use $\alpha = 0.1$.

1. Find the four predictions and the four residuals.
2. Compute $\dfrac{\partial L}{\partial m}$ and $\dfrac{\partial L}{\partial b}$.
3. Take one gradient descent step. What are the new m and b?
4. Compute the loss before the step. (Bonus: after it.)

Exercise: Solution

Answers

1. Predictions $1,\ 1.5,\ 2,\ 2.5$; residuals $e_i = 0,\ 1.5,\ 0,\ 1.5$.
2. $\sum x_i e_i = 0 + 3 + 0 + 6 = 9$ and $\sum e_i = 3$, so
$$\frac{\partial L}{\partial m} = -\tfrac{2}{4}(9) = -4.5, \qquad \frac{\partial L}{\partial b} = -\tfrac{2}{4}(3) = -1.5.$$
3. $m = 0.5 - 0.1(-4.5) = 0.95$ and $b = 0.5 - 0.1(-1.5) = 0.65$.
4. Before: $L = \frac{1}{4}(0 + 2.25 + 0 + 2.25) = 1.125$.
After: residuals $-0.6,\ 0.45,\ -1.5,\ -0.45$, so $L = \frac{1}{4}(3.015) \approx 0.754$.

Meaning: the start line sat below every point it missed, so the step raised both m and b and the loss fell. m overshot 0.8; later steps pull it back, the zigzag gradient descent makes in a narrow valley.

Review: Optimization and Linear Regression

What we covered: finding extrema with calculus

1. Optimization finds the min or max of an **objective function** f over a **search space** D. The **minimizer** x^* says where; the **minimum value** $f(x^*)$ says how good. A max of f is a min of $-f$.
2. A **global** minimum is lowest over all of D; a **local** one is lowest only nearby. Algorithms can mistake one for the other, and with many inputs we cannot plot to check.
3. **Derivative test**: at a smooth extremum inside D, $f'(x) = 0$. Solve for the critical points, add any endpoints, and compare values.
4. $f'(x) = 0$ gives candidates, not answers (x^3 at 0). With a restricted D, the best value can sit at an endpoint where $f' \neq 0$.
5. f'', the derivative of f', measures bending: $f'' > 0$ concave up (cup), $f'' < 0$ concave down (cap). At a critical point, $f'' > 0$ means local min, $f'' < 0$ local max, and $f'' = 0$ says nothing.

Review: Optimization and Linear Regression

What we covered: gradient descent

6. When $f'(x) = 0$ cannot be solved, **gradient descent** searches instead: $x_{k+1} = x_k - \alpha f'(x_k)$. Step against the slope, and repeat. It only ever evaluates f'.
7. Steps shrink on their own as the slope flattens; at a minimum the walk stops.
8. The **learning rate** α: too small crawls, too big overshoots, bounces, or blows up.
9. Stop after N steps, or when the slope or the step falls below a tolerance.
10. Gradient descent finds **a** local minimum, and which one depends on the start. Restarts help.

Review: Optimization and Linear Regression

What we covered: the gradient

11. The **gradient** $\nabla f = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \end{bmatrix}$ points steepest uphill, crosses contours at right angles, and its length is the steepness. $\nabla f = \bar{0}$ at flat points.
12. The slope in the direction $\hat{u}$ is $\hat{u} \cdot \nabla f = \|\nabla f\| \cos\theta$.
13. **Gradient descent** in the plane: $\begin{bmatrix} x_{k+1} \\ y_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \end{bmatrix} - \alpha \nabla f(x_k, y_k)$.

Review: Optimization and Linear Regression

What we covered: the best line

14. The model $\hat{y} = mx + b$; residual $e_i = y_i - \hat{y}_i$; loss $L(m, b) = \frac{1}{n} \sum e_i^2$. Squaring stops signs cancelling and makes big misses cost more.
15. The data are fixed, so L is a function of m and b: a **bowl** whose one lowest point is the best line.
16. **Way 1**: set $\frac{\partial L}{\partial m} = \frac{\partial L}{\partial b} = 0$ to get the normal equations, a 2×2 linear system. In matrix form, $\left[\begin{smallmatrix} m \\ b \end{smallmatrix}\right] = (X^\top X)^{-1} X^\top \bar{y}$.
17. **Way 2**: gradient descent on L reaches the same line. It still works when no formula exists, which is why it trains every neural network.
18. More features: one more coefficient and one more partial each. Nothing else changes.

Notebooks: `U3-5_Optimize-1`; `U3-3_Derivatives-3`; `U3-6_Multivariable-3`; `U3-7_GradientDescent-1`; `U3-7_Regression-1` to `-6`. **Previous:** Unit 3-1, Limits and Derivatives. **Next:** Unit 3-3, **Integrals**.